

Ma, Hao

Max Planck Institute for Intelligent Systems (MPI-IS)
Eidgenössische Technische Hochschule Zürich (ETHZ)
Center for Learning Systems (CLS)

☎ +41-772838467
✉ hao.ma@tuebingen.mpg.de
✉ haomah@ethz.ch
G Google Scholar
G Github
in LinkedIn
🔗 Personal Profile

ABOUT ME

My research focuses on *embodied intelligence*, with the goal of enabling robots to achieve autonomous learning in the real world. I believe this cannot be achieved through a single approach, but instead calls for a system-level perspective that brings together adaptive control, safety-aware planning, and on-device intelligence. I approach these challenges through model-based online learning, constraint-aware generative planning, and efficient inference of foundation models on edge devices.

EDUCATION

- **ETH Zurich, Switzerland** 04.2022 – 03.2026
Doctorate of Mechanical Engineering
- **Technical University of Munich, Germany** 04.2019 – 11.2021
Master of Automotive Engineering GPA: 1.1/1.0
- **Karlsruhe Institute of Technology, Germany** 10.2018 – 04.2019
Master of Mechatronics and Information Technology
- **Jilin University, China** 09.2013 – 07.2017
Bachelor of Energy and Power Engineering (Automotive Engine) GPA: 3.13/4.0

RESEARCH EXPERIENCE

- **ETH Zurich, Switzerland** 04.2026 – Now
Postdoctoral Researcher
- **Max Planck Institute for Intelligent Systems, Germany** 04.2022 – Now
Scientific Researcher

PUBLICATIONS

- **Hao Ma**, Melis Ilayda Bal, Liang Zhang, Bingcong Li, Niao He, Melanie Zeilinger, Michael Muehlebach. SALAAD: Sparse And Low-Rank Adaptation via ADMM for Large Language Model Inference. *International Conference on Machine Learning*, pages 1-18, 2026.
- Fang Nan, **Hao Ma**, Qinghua Guan, Josie Hughes, Michael Muehlebach and Marco Hutter. Efficient Model-Based Reinforcement Learning for Robot Control via Online Learning. *arXiv*, pages 1-16, 2025.
- **Hao Ma**, Sabrina Bodmer, Andrea Carron, Melanie Zeilinger and Michael Muehlebach. Constraint-Aware Diffusion Guidance for Robotics: Real-Time Obstacle Avoidance for Autonomous Racing. *Conference on Robot Learning*, pages 1-19, 2025.
- **Hao Ma**, Melanie Zeilinger and Michael Muehlebach. Online Optimization of Closed-Loop Control Systems. *ICML Workshop*, pages 1-16, 2024.
- Simon Guist, Jan Schneider, **Hao Ma**, Le Chen, Vincent Berenz, Julian Martus, Heiko Ott, Felix Grüninger, Michael Muehlebach, Jonathan Fiene, Bernhard Schölkopf and Dieter Buehler. Safe & Accurate at Speed with Tendons: A Robot Arm for Exploring Dynamic Motion. In *Proceedings of Robotics: Science and Systems*, pages 1-12, 2024.
- **Hao Ma**, Melanie Zeilinger and Michael Muehlebach. Stochastic Online Optimization for Cyber-Physical and Robotic Systems. *Machine Learning*, pages 1-46, 2024.

- **Hao Ma**, Dieter Büchler, Bernhard Schölkopf and Michael Muehlebach. Reinforcement learning with model-based feedforward inputs for robotic table tennis. *Autonomous Robots*, 47(8):1387-1403, 2023.
- Philip Tobuschat, **Hao Ma**, Dieter Büchler, Bernhard Schölkopf and Michael Muehlebach. Data-Efficient Online Learning of Ball Placement in Robot Table Tennis. In *Proceedings of International Conference on Intelligent Robots and Systems*, pages 567-573, 2023.
- Jan Achterhold, Philip Tobuschat, **Hao Ma**, Dieter Büchler, Michael Muehlebach and Joerg Stueckler. Black-Box vs. Grey-Box: A Case Study on Learning Table Tennis Ball Trajectory Prediction with Spin and Impacts. In *Proceedings of Learning for Dynamics and Control Conference*, pages 878-890, 2023.
- Simon Guist, Jan Schneider, **Hao Ma**, Vincent Berenz, Julian Martus, Felix Grüninger, Michael Muehlebach, Jonathan Fiene, Bernhard Schölkopf and Dieter Büchler. A Robust Open-source Tendon-driven Robot Arm for Learning Control of Dynamic Motions. In *Proceedings of RoboLetics: Workshop on Robot Learning in Athletics*, pages 1-3, 2023.
- **Hao Ma**, Dieter Büchler, Bernhard Schölkopf and Michael Muehlebach. A Learning-based Iterative Control Framework for Controlling a Robot Arm with Pneumatic Artificial Muscles. *Robotics: Science and Systems*, pages 1-10, 2022.

TEACHING EXPERIENCE

- **Ashesi University** 02.2026 (block course)
Lecturer Ghana
– I participated in the ETH–Ashesi MSc/MAS Mechatronics program at Ashesi University in Ghana, where I taught the course “Signals and Systems” and was primarily responsible for delivering the lectures.
- **Eidgenössische Technische Hochschule Zürich** 06.2024 (block course) and 02.2023 – 06.2023
Teaching Assistant Switzerland
– I served twice as a teaching assistant for Dr. Michael Muehlebach at Eidgenössische Technische Hochschule Zürich for the course “Large-Scale Convex Optimization”. I was mainly responsible for the colloquia and designing exercises and exams.
- **Max Planck Institute for Intelligent Systems** 05.2021 – 03.2022
Scientific Researcher Germany
– I did an internship at the Learning and Dynamical Systems Group at the Max Planck Institute for Intelligent Systems under the supervision of Dr. Michael Muehlebach.

AWARDS

- **Max-ETH Center for Learning Systems Fellowship** 05.2022 – 05.2024
highly competitive fellowship funding Ph.D. studies (acceptance rate 3% – 4%)
- **Deutschlandstipendium** 10.2020 – 09.2021
awarded by the Bundesministerium für Bildung und Forschung
- **Third-class Scholarship** 09.2015 – 07.2016
awarded by Jilin University
- **Third-class Scholarship** 09.2014 – 07.2015
awarded by Jilin University
- **Second-class Scholarship** 09.2013 – 07.2014
awarded by Jilin University

PUBLIC COMMUNICATION

•Talk at the EWRL 2025

2025

European Workshop on Reinforcement Learning

- I presented a contributed talk at the EWRL 2025, entitled *Provably Efficient Online Learning in Real-World Cyber-Physical and Robotic Systems*.

•Wissen Was mit Doktor Watson

2024

Youtube video with science communicator Cedric Engels

- I supported the live demonstration of a ping-pong playing robot during my supervisor's media interview.

•Max Planck Research Magazine

2023

Special issue on "Orientation"

- I supported the live demonstration and explanation of a ping-pong playing robot featured in my supervisor's magazine interview.

•Science and Innovation Days Tübingen

2023

Live robotics presentation to the general public in Tübingen

- I delivered a live presentation on the working principles of a ping-pong playing robot, showcasing its functionality and enabling hands-on interaction for the public.

TECHNICAL SKILLS

Programming Languages: Matlab/Simulink, Python

Frameworks & Libraries: ROS, PyTorch, TensorFlow, MuJoCo, Isaac Gym, Brax, Hugging Face

Tools & Platforms: Git, Docker, VSCode, Jupyter, HPC Clusters

Operating Systems: Windows, Linux

Document Creation: Microsoft Office Suite, LaTeX, TikZ

Language Skills: Chinese(Native), English(C1), German(DSH-2)